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GEOGRAPHY - 11

World Climate and Climate Change

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World Climate and Climate Change

Introduction:

The term "**climate**" originates from the **Greek word "Klima,"** which signifies inclination, slope, or latitude. Climate refers to the average meteorological conditions at a particular location over an extended period. The study of world climate involves organizing climate data and information into smaller, more manageable units for better understanding, description, and analysis.

Factors Influencing Climate:

- Latitudinal and longitudinal extent
- Sea level
- Distance from continents and oceans
- Prevalent wind types
- Precipitation patterns
- Oceanic waves and currents
- Relief features
- Soil type
- Local geographic characteristics

Classification of Climate:

The **Greeks pioneered the classification of climate** based on **temperature**, distinguishing between **warm, temperate and cold zones**, which gradually extend from the **Equator to the Poles**. The continent of Asia spans from the Equator to the North Pole, encompassing diverse climate zones. Regions with consistent climates are termed climatic regions. Key factors in determining climate include **latitude, altitude, proximity to the sea, temperature, vegetation types**, and the **impact of evaporation**. Among these, latitude is the primary determinant of an area's climate.

Approaches for Classifying Climate:

Empirical Classification:

- This approach is **based on observed data**, particularly focusing on measurable variables such as temperature and precipitation.
- **Example – Koeppen Climate Classification**, which uses temperature and precipitation data to categorize different climate types

Genetic Classification:

- This method classifies climates according to the **causes or origins** of their climatic conditions.
- **Example - Classification based on air mass** analysis or the dynamics of atmospheric circulation patterns.

Applied Classification:

- This approach is tailored for **specific purposes or applications**, such as agricultural suitability or tourism.
- **Example - Classifying climates based on their suitability for growing particular crops or for tourism activities.**

Koeppen's Scheme of Classification of Climate:

The **most widely used** classification of climate is the empirical scheme developed by **V. Koeppen**. He identified a **close relationship between the distribution of vegetation and climate**. He selected certain values of **temperature and precipitation** and related them to the distribution of vegetation and used these values for classifying the climates.

Basis of Classification:

- Empirical classification based on **mean annual and mean monthly temperature** and **precipitation** data.

Designation of Climatic Groups:

- Use of **capital and small letters** to designate **climatic groups and types**.
- Five major climatic groups - 4 based on temperature (A, C, D, E) and 1 on precipitation (B).**
- The climatic groups are **subdivided into types**, designated by **small letters**, based on seasonality of precipitation and temperature characteristics.

Climatic Group Designations:

- The **seasons of dryness** are indicated by the **small letters (f, m, w and s)**
 - f** - no dry season.
 - m** - monsoon climate.
 - w** - winter dry season.
 - s** - summer dry season.
- Small letters - a, b, c, and d** refer to the degree of **severity of temperature**.
- The **B - Dry Climates** are subdivided using the capital letters **S** for **steppe** or semi-arid and **W** for **deserts**.

Table 11.2 : Climatic Types According to Koeppen

Group	Type	Letter Code	Characteristics
A-Tropical Humid Climate	Tropical wet	Af	No dry season
	Tropical monsoon	Am	Monsoonal, short dry season
	Tropical wet and dry	Aw	Winter dry season
B-Dry Climate	Subtropical steppe	BSh	Low-latitude semi arid or dry
	Subtropical desert	BWh	Low-latitude arid or dry
	Mid-latitude steppe	BSk	Mid-latitude semi arid or dry
	Mid-latitude desert	BWk	Mid-latitude arid or dry
C-Warm temperate (Mid-latitude) Climates	Humid subtropical	Cfa	No dry season, warm summer
	Mediterranean	Cs	Dry hot summer
	Marine west coast	Cfb	No dry season, warm and cool summer
D-Cold Snow-forest Climates	Humid continental	Df	No dry season, severe winter
	Subarctic	Dw	Winter dry and very severe
E-Cold Climates	Tundra	ET	No true summer
	Polar ice cap	EF	Perennial ice
H-Highland	Highland	H	Highland with snow cover

Table 11.1 : Climatic Groups According to Koeppen

Group	Characteristics
A - Tropical	Average temperature of the coldest month is 18 °C or higher
B - Dry Climates	Potential evaporation exceeds precipitation
C - Warm Temperate	The average temperature of the coldest month of the (Mid-latitude) climates years is higher than minus 3 °C but below 18 °C
D - Cold Snow Forest Climates	The average temperature of the coldest month is minus 3 °C or below
E - Cold Climates	Average temperature for all months is below 10 °C
H - High Land	Cold due to elevation

Tropical Humid Climates (Group A):

Regions:

These climates exist between the **Tropic of Cancer** and **Tropic of Capricorn**.

Characteristics:

The **sun is overhead throughout the year**, and the presence of the **Inter Tropical Convergence Zone (ITCZ)** makes the **climate hot and humid**. **Low annual range of temperature** and **high annual rainfall**.

The tropical group is **divided into 3 subtypes** – Tropical wet climate (**Af**), Tropical monsoon climate (**Am**), Tropical wet and dry climate (**Aw**).

Tropical Wet Climate (Af):

- **Regions** - Near the equator (Amazon Basin, western equatorial Africa, East Indies).
- **Rainfall** - Significant rainfall every month, typically as afternoon thunderstorms.
- **Temperature** - Uniformly high (maximum around 30°C, minimum around 20°C).
- **Vegetation** - Tropical evergreen forests with dense canopy and high biodiversity.

Tropical Monsoon Climate (Am):

- **Regions** - Indian subcontinent, northeastern South America, northern Australia.
- **Rainfall** - Heavy in summer; dry winter.
- **Vegetation** - Monsoon forests with trees shedding leaves during the dry period; high biodiversity.

Tropical Wet and Dry Climate (Aw):

- **Regions** - North and south of Af climate regions, bordering dry climates (western continents) and Cf/Cw climates (eastern continents).
- **Rainfall** - Less than Af and Am; shorter wet season, longer dry season with severe drought.
- **Temperature** - High throughout the year; greatest diurnal range in the dry season.
- **Vegetation** - Deciduous forests and tree-shredded grasslands.

Dry Climates (Group B):

Regions:

Large areas from **15° to 60° north and south of the equator**. At **low latitudes (15°- 30°)**, associated with **subtropical highs**. At **middle latitudes (35°- 60°)**, interior continents where maritime winds do not reach.

Characteristics:

Dry climates are characterized more by the **lack of precipitation** than by temperature. In these regions, **evaporation and transpiration exceed the total precipitation**.

Dry climates are **divided into** Steppe or Semi-arid climate (**BS**) and Desert climate (**BW**).

Dry Arid Climate or True Desert Climate (BW):

- **Rainfall** - Extremely low, often **less than 25 cm annually**. The dry conditions support **xerophytic vegetation**, which is adapted to conserve water.
- **Temperature** - These areas experience **extreme temperature variations** between **day and night**. **Maximum temperature** in the summer is very high. The highest shade temperature of **58° C** was recorded at **Al Aziziyah, Libya** on 13 September 1922. The **annual and diurnal ranges** of temperature are also **high**.
- **Subtypes** -
 - **BWh** - Subtropical deserts, such as the **Sahara**.
 - **BWk** - Mid-latitude deserts, such as the **Gobi Desert**.

Dry Semi-Arid Climate or Steppe Climate B(S):

- **Rainfall** - Higher than in deserts, ranging **from 25 to 50 cm annually**, often due to **mid-latitude cyclones** or the inter-tropical convergence zone.
- **Temperature** - Milder than BW regions, but still with significant temperature variations.
- **Vegetation** - Supports **grasslands** rather than deserts.
- **Subtypes** -
 - **BSh** - Subtropical steppes, such as parts of the **Sahel**.
 - **BSk** - Mid-latitude steppes, such as the **Great Plains in the USA**.

Warm Temperate (Mid-Latitude) Climates (Group C):

Regions:

Moist sub-tropical mid-latitude climates are found between **30 to 50 degrees latitude** north and south of the equator. These climates are mainly located on the **eastern and western margins of continents**.

Characteristics:

Moist sub-tropical mid-latitude climates feature **warm, humid summers and mild winters**.

Humid Subtropical Climate (Cwa):

- **Regions** - Occurs **poleward of the Tropic of Cancer and Capricorn** (North Indian plains, South China interior plains).
- **Characteristics** - Similar to Aw climate but with **warm winters**.

Mediterranean Climate (Cs):

- **Regions** - Occurs around the **Mediterranean Sea** and on the **west coast of continents in subtropical latitudes (30°-40°)**. Influenced by **subtropical high** in **summer** and **westerly winds** in **winter**.
- **Characteristics** - **Hot, dry summers** and **mild, rainy winters**. Summer temperatures around 25°C; winter below 10°C. **Annual precipitation ranges from 35-90 cm**.

Humid Subtropical Climate (Cfa):

- **Regions** - Located in eastern parts of continents (eastern USA, southern and eastern China, southern Japan, northeastern Argentina, coastal South Africa, eastern coast of Australia).
- **Rainfall** - Rainfall **throughout the year**. **Thunderstorms** in **summer** and **frontal precipitation** in **winter**.
- **Temperature** - Summer mean around 27°C; winter between 5°C - 12°C.

Marine West Coast Climate (Cfb):

- **Regions** - **Poleward of the Mediterranean climate** on the **west coast of continents** (northwestern Europe, west coast of North America, southern Chile, southeastern Australia, New Zealand).
- **Temperature** - Moderate; summer 15°C - 20°C, winter 4°C - 10°C.
- **Rainfall** - Occurs year-round; ranges from 50 - 250 cm.

Cold Snow Forest Climates (Group D):

Regions:

Located in **large continental areas** in the **northern hemisphere** between **40°-70° north latitudes** (Europe, Asia, North America).

Characteristics:

The **warmest month** averages above **10° C**, and the **coldest month** averages below **3° C**. **Winters are severe**, especially in higher latitudes. Harsh, **with strong winds** and frequent **snowstorms**.

Cold Climate with Humid Winters (Df):

- **Regions** - Poleward of **marine west coast** climate and mid-latitude steppe.
- **Characteristics** - **Cold, snowy winters**; short frost-free season; **large annual temperature ranges**.

Cold Climate with Dry Winters (Dw):

- **Regions** - Mainly in **northeastern Asia**.
- **Characteristics** - Winter **anti-cyclones** and summer weakening cause monsoon-like wind reversals.
- **Temperature** - **Low summer temperatures; extremely cold winters**, below freezing for up to seven months.
- **Rainfall** - Occurs in **summer**; annual precipitation low (12 - 15 cm).

Polar Climates (Group E):

Regions:

Northernmost parts of Asia, northernmost parts of Europe, Northernmost parts of North America, Greenland and Antarctica.

Characteristics:

Extremely low temperatures throughout the year. The **warmest month** averages below **10° C**.

Tundra Climate (ET):

- **Vegetation** - Low-growing **mosses, lichens**, flowering plants.
- **Permafrost** - Subsoil permanently frozen.
- **Summer** - Long daylight duration; supports low-growing plants.

Ice Cap Climate (EF):

- **Regions** - Found in interior Greenland and Antarctica.
- **Temperature** - Below freezing even in summer.
- **Rainfall** - Very little; snow and ice accumulate, form icebergs that float in polar waters.

Highland Climates (H):

Governed by **topography**, with **significant temperature changes over short distances**. Precipitation types and intensity vary spatially, leading to vertical zonation of climatic types with **elevation in mountainous environments**.

Pros and Cons of Koeppen Classification:

Pros:

- **Quantitative** - The classification system is based on measurable data, making it easier to understand and apply.
- **Vegetation Correlation** - It aligns well with global vegetation patterns, providing a useful ecological context.
- **Effective Precipitation** - Incorporates the concept of evapotranspiration, which is crucial for understanding water balance in different climates.

Cons:

- **Average Values Emphasis** - Overemphasis on mean monthly temperature and precipitation values, which can obscure important climatic details.
- **Ignored Factors** - Does not consider important weather elements such as precipitation intensity, cloud cover, daily temperature variations, and the number of rainy days.
- **Air Masses Role** - Fails to account for the influence of air masses, which are fundamental to modern climatology.
- **Empirical Basis** - Being empirical, it relies on observed data rather than underlying climatic processes, which limits its applicability in explaining climate dynamics.

Climate Change:

The planet earth has witnessed many variations in climate since beginning. There are various evidence showing that **climate change is natural and continuous process**. **Geological records** show alteration of glacial and inter-glacial periods. The **geomorphological features**, especially in high altitudes and high latitudes, exhibit traces of advances and retreats of glaciers. The **sediment deposits in glacial lakes** also reveal the occurrence of warm and cold periods. **The rings in the trees** provide clues about wet and dry periods. **The Rajasthan desert experienced wet and cool climate around 8,000 B.C.**

Stages of Climate Change:

In the geological past, the **earth was warm some 500-300 million years ago**, through the **Cambrian, Ordovician and Silurian periods**. During the **Pleistocene epoch**, glacial and inter-glacial periods occurred, the **last major peak glacial period** was about **18,000 years ago**. The **present inter-glacial period** started **10,000 years ago**.

Causes of Climate change:

Climate change can be attributed to a variety of factors, which can be broadly classified into **astronomical and terrestrial causes**.

Astronomical Causes:

- **Changes in Solar Output** - The variation in solar output is linked to **sunspot activities**. **Sunspots are dark, cooler patches on the sun that increase and decrease cyclically**. An **increase in sunspots** is believed by some meteorologists to cause **cooler, wetter weather and greater storminess**, while a **decrease in sunspots** is associated with **warmer, drier conditions**. However, these findings lack statistical significance.
- **Milankovitch Oscillations** - This theory involves **cycles in the earth's orbital characteristics around the sun**, including the wobbling of the earth and changes in its **axial tilt**. These variations **alter the amount of solar radiation** (insolation) received by the Earth, potentially impacting the climate.

Terrestrial Causes:

- **Volcanism** - **Volcanic eruptions release large amounts of aerosols** into the atmosphere. These aerosols can remain in the atmosphere for extended periods, **reducing the sun's radiation that reaches the Earth's surface**.
 - **Example – The Pinatubo and El Chichon volcanic eruptions** led to a slight decrease in the average global temperature for several years.
- **Anthropogenic Effects** - The **most significant human-induced effect** on the climate is the **increasing concentration of greenhouse gases** in the atmosphere, which is likely to result in **global warming**.

Impacts of Climate Change:

Environmental Impacts:

- **Rising Temperatures** - Increased average global temperatures lead to more frequent and severe heatwaves. Changes in weather patterns and seasons, affecting ecosystems and biodiversity.
- **Melting Ice and Rising Sea Levels** - Glacial retreat and the melting of polar ice caps contribute to rising sea levels. Increased coastal erosion and higher risk of flooding in low-lying areas.

- **Ocean Acidification** - Increased carbon dioxide absorption by oceans leads to acid
- **Extreme Weather Events** - Increased frequency and intensity of hurricanes, typhoons, and cyclones, as well as more severe droughts and heavy rainfall events leading to floods.
- **Disruption of Ecosystems** - Altered habitats and migration patterns for wildlife, with an increased risk of extinction for many species unable to adapt to rapid changes.
- **Changes in Agricultural Productivity** - Shifts in growing seasons and agricultural zones, and increased risk of crop failures due to extreme weather and changing precipitation patterns.

Human Health Impacts:

- **Heat-Related Illnesses** - Increased incidence of heat strokes and dehydration, with higher mortality rates during extreme heat events, particularly among vulnerable populations.
- **Vector-Borne Diseases** - Expanded range of diseases like malaria, dengue, and Zika due to changing habitats for mosquitoes and other vectors, leading to increased public health challenges in controlling outbreaks.
- **Food Security** - Reduced crop yields and increased food prices, creating a greater risk of malnutrition and food shortages, particularly in developing countries.

Economic Impacts:

- **Damage to Infrastructure** - Increased costs of repairing and replacing infrastructure damaged by extreme weather events, along with higher insurance premiums and increased economic losses.
- **Impacts on Livelihoods** - Disruption of industries such as agriculture, fishing, and tourism, leading to increased poverty and displacement as communities lose their livelihoods.

Social Impacts:

- **Climate Migration** - Forced displacement of populations due to sea-level rise, extreme weather, and environmental degradation, resulting in increased pressure on urban areas and potential for conflict over resources.
- **Water Scarcity** - Changes in precipitation patterns leading to water shortages, with increased competition for water resources among agriculture, industry, and domestic use.

Political and Security Impacts:

- **Resource Conflicts** - Increased potential for conflicts over diminishing natural resources, such as water and arable land, and greater geopolitical tensions as nations compete for resources and deal with climate-induced migration.

Cultural Impacts:

- **Loss of Heritage** - Threats to cultural heritage sites from rising sea levels and extreme weather, leading to the disruption of traditional ways of life and indigenous practices.

Global Impacts:

- **Impact on Biodiversity** - Significant loss of biodiversity affecting ecosystems' resilience and services, disrupting global ecological balance and functions.

Global Warming:

Global warming is caused by **greenhouse gases (GHGs) in the atmosphere**, which **trap heat and cause the greenhouse effect**. These gases **absorb most of the long-wave radiation** emitted upwards from the Earth's surface, leading to a rise in global temperatures.

Greenhouse Gases (GHGs):

- **Carbon Dioxide (CO₂)**
- **Chlorofluorocarbons (CFCs)**
- **Methane (CH₄)**
- **Nitrous Oxide (N₂O)**
- **Ozone (O₃)**

Other gases, such as **nitric oxide (NO)** and **carbon monoxide (CO)**, can interact with GHGs, affecting their concentrations in the atmosphere.

The Effectiveness of Any GHG Depends On:

- The **magnitude of the increase in its concentration**.
- Its **lifetime in the atmosphere**.
- The **wavelength of radiation it absorbs**.

CFCs are highly effective, while ozone, which absorbs ultraviolet radiation in the stratosphere, is also significant when present in the lower troposphere.

Carbon Dioxide (CO₂) - Carbon dioxide is the **most prevalent greenhouse gas**. It is primarily emitted through the combustion of fossil fuels such as oil, gas, and coal. **Forests and oceans act as sinks for CO₂**, with forests using CO₂ for growth. **Deforestation** increases CO₂ concentrations by reducing these sinks. CO₂ takes 20-50 years to adjust to changes in sources and sinks, and its concentration is rising by about 0.5% annually. Climate models use a doubling of CO₂ concentration from pre-industrial levels to estimate climate changes.

Chlorofluorocarbons (CFCs) - CFCs are **human-made compounds** that contribute to the **destruction of ozone** in the stratosphere. Ozone absorbs ultraviolet rays, protecting the Earth from harmful radiation. CFCs, however, degrade ozone, leading to significant depletion over Antarctica and the formation of the "ozone hole," allowing more ultraviolet rays to reach the Earth's surface.

International Efforts:

- The **Kyoto Protocol**, adopted in **1997** and effective from 2005, is a key international initiative to reduce GHG emissions. It committed 35 industrialized countries to **reduce their emissions by 5% below 1990 levels by 2012**.

Impacts of Global Warming:

- Global warming has serious consequences for life-supporting systems. Rising sea levels due to melting glaciers and thermal expansion threaten coastal areas and islands. The trend of increasing GHG concentrations and global temperatures, with the seven warmest years of the 1856-2000 period occurring in the last decade of the 20th century, underscores the urgency of addressing this issue. The year 1998 was noted as the warmest year of the 20th century, and possibly the warmest of the millennium.
- Efforts to control GHG emissions and mitigate global warming are crucial. Global cooperation is essential to address these challenges and ensure a liveable world for future generations.